Blessed Gebriel Catholic Secondary School

2nd Semester Mathematics test one for Grade 12th

March, 2018 E.C

Time Allowed 1:30

Name:_____ Sec:_____ ID No._____

I. Choose the best Answer from the give alternative.

1. Which one of the following true bout the given 2,4,6

A. mean = 3 ✓ B. variance = 2.67 ✓ C. median = 2 ✗ D. standard deviation 2 ✗

2. The mean deviation of 2,4,6,8 is _____

A. 4 B. 3 C. 2 D. 6

3. The quartile deviation of 2,4,6,8,10 is _____

A. 4 B. 2 C. 6 D. 3

4. The coefficient of quartile deviation if Q1= 10 and Q3 = 20 is _____

A. 0.33 B. 0.6 C. 0.4 D. 0.48

5. Which one of the following is the interquartile range of these data sets of

x	0-5	6-10	11-15	16-20
f	2	1	3	2

A. 10 B. 11 C. 14 D. 6

6. Which one of the following is mean deviation about the mean of 4,6,8,10 is _____

A. 2 B. 3 C. 4 D. 1

7. Which one of the following is mean deviation about the mode of 4 ,6,6,8,10 is _____

A. 4 B. $\frac{8}{5}$ C. $\frac{3}{4}$ D. $\frac{5}{3}$

8. If the data

x	0-9	10-19	20-29	30-39
f	1	2	3	2

Which one of the following is the median of the data

A. 20 B. 22.8 C. 14 D. 14.2

9. Which one of the following is the solution set of the inequality of $y+\frac{1}{2}x \geq 3$

B. A. (0,1) ✗ B. (1,2) ✗ C. (4,3) ✓ D. (-2,2)

10. The coefficient of mean deviation about the mean of

X	0	1	2	3	4
f	1	9	7	3	4

is:-

A. 0.46 B. 96 C. 46 D. 16

11. Which one of the following is absolute measures of dispersion

A. mean deviation

B. coefficient of range

C. coefficient measure of dispersion

D. coefficient of standard deviations

12. For the given data set

x	4	5	6	8
f	1	2	4	2

The distribution pearson's coefficient is _____

A. Symmetrical B. skewed to the right C. skewed to the left D. skewed to left to right

13. The mean and coefficient of variation of seven observation are 8 and 50, respectively. If five of those are 12,14,10,4 and 2 then the remaining two observations are _____ and _____

A. 6 and 8 B. 4 and 6 C. 10 and 16 D. 8 and 5

14. The mean and standard deviation of 15 observations are found to be 10 and 5, respectively. On rechecking it was found that one of the observations with the value 8 was incorrect. The mean and standard deviation if the value of the corrected observation was 21 is _____?

A. 10 B. $\sqrt{35}$ C. $\sqrt{45}$ D. $\sqrt{135}$

15. The solution set of the system $4(2x-8) < 2x+4$

A. $x<4$ B. $x>4$ C. $x<6$ D. $x>7$

16. Let $\begin{cases} 4y+x \geq 10 \\ 2y-x < 6 \end{cases}$, then which one of the following is the solution set of the system

A. (1,2) B. (4,3) C. (1,4) D. (0,1)

17. Which one of the following is the maximum value of linear programming problem of for $Z= x+y$, subject to

$$\begin{cases} 2y+x \leq 4 \\ x-1 \leq y \\ x \geq 0 \\ y \geq 0 \end{cases}$$ is _____

A. 3 B. 4 C. 6 D. 5

18. Which one of the following is the value of x and y that maximize $Z= x+3y$

$$\text{Subject to} \begin{cases} 2x-24 \leq -3y \\ x-y < 7 \\ y \leq 6 \\ x \geq 0 \\ y \geq 0 \end{cases}$$ is _____

A. (3,6) B. (9,2) C. (1,2) D. (4,6)

II. Short answer

19. A company is making two products A and B. The cost of producing one unit of product A and B is Birr 60 and Birr 80 respectively. As per the agreement, the company has to supply at least 200 units of product B to its regular customers per month. One units of product A requires one machine hours where as product B has machine hours a valuable abundantly within the company. Total machine hours available for product A are 400 hours. One unit of each product A and B required one labor hour each and total of 500 labor hours are available. The company wants to minimize the cost of production by satisfying the given requirement formulate the problem as a linear programming problem and find the minimum cost.

20. A carpenter's daily production of chairs increased from 30 units to 50 units. At the same time his daily income increased 1600 Birr to 2400 Birr. As a result of this increase what is the rate of change of income per unit is _____.

21. 35% of 1500 is _____

22. The monthly salary of Mr. Abebe is 20% more than Mr. Tilahun salary . Assume Mr. Tilahun salary is 20,000. What is the salary of Mr. Abebe.

23. Calculate the regular price of a car assuming that 20% discount would have been worth Birr 4000 is _____

24. Aster loaned Awoke Birr 10,000 for 6 months Awoke paid back Birr 10,400. How much interest did he pay? What is the annual interest rate?

25. Find the term of a loan if the principal is Birr 15000, the interest rate 5% and the total interest is Birr 300 is _____

✓26. How much does kedija need in her investment account if she wants to be able to receive Birr 450 per month in income form it. Assuming a simple interest rate of 7.5% of annually?

27. How much silver is found is 30kg of an alloy that 5% of silver?

✓28. The price of cement is Birr 2500 per quintal. If you sell it for Birr 2800 per quintal, what is the mark up per quintal?

✓29. Solve $\frac{2x}{3} - 3 = \frac{16}{21} - \frac{13}{3} - \frac{2x}{15}$ is _____

30. A book costs 400 birr and is marked up by 20% .If a discount of 10% is given, find the profit.

31. A trader buys a bag for 600 Birr, marks it up by 25%, then gives a discount of 10%. Find the final selling price.

Bonus

An item is sold for 540 birr after a 10% discount. If the mark up was 20% .find the cost price.

Blessed G/Michael Catholic Secondary School

2nd Semester Mathematics test one for Grade 12th

March, 2018 E.C

Time Allowed 1:30'

Name: Hanna Ayalew Sec: 12th C R.No. 21

1. B	6. A	11. A	16. B
2. C	7. B	12. A	17. A
3. A	8. B	13. A	18. D
4. A	9. C	14. B	
5. B	10. A	15. C	

(19) *

20. 10

21. 525

22. 24,000

23. 20,000

24. I = 400.
r = 8%

25. t = 3/5 years

26. = 400 birr

27. 1.5 kg.

28. 300

29. -5/7

30, 32

31. 675

Bonus

$32.\ 540 = S.P - S.P\left(\frac{10}{100}\right)$ (2)

$540 = 0.9S.P$

$SP = 600.$

$20\% = \frac{600 - C.P}{C.P} * 100\%$

$20CP = 60000 - 100C.P.$

$120C.P = 60000$

$C.P = 500$

(19) A - x
B - y

$Z = 60x + 80y$

$y \geq 200$
$x \leq 400.$
$x + y \leq 500$
$x \geq 0$
$y \geq 0$

minimum cost at A = 0
B = 200

$Z = 60(0) + 80(200)$
$= 16000$